MARCO ANTONIO FAGANELLO

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SUMMARY & SKILLS

Consultant and Data Scientist with a PhD in Political Science (UNICAMP) and a Postdoctoral fellowship at FGV, specializing in applying quantitative methods to generate insights on public policy, electoral dynamics, and social challenges. I lead end-to-end data projects – from data extraction from restricted sources (web scraping, APIs) and structuring large-scale datasets to applying machine learning and econometric models to inform strategic decision-making.

Core Competencies: Python | R | Redes Neurais | Machine Learning | Inteligência Artificial | BERT | GIS | Dados Quantitativos | Análise Causal (RDD, DiD, Regressão) | Web Scraping | Gephi | Mapbox | Kotlin | Javascript | Sala de Sigilo | CNEFE | TSE | IBGE | xlsform

EDUCATION

UNICAMP – Campinas/SP December 2022

PhD in Political Science | Master's in Political Science

My hybrid background in data science and the humanities culminated in a doctoral dissertation that applied econometrics to investigate the causes of party fragmentation in Brazil. My expertise covers the application of quantitative methods to the analysis of public policy, electoral systems, and voting dynamics, including its behavior and geography.

MIT Political Science – Boston, MA July 2020

Visiting Scholar

- Participated in an international research program at the Massachusetts Institute of Technology (MIT), focusing on advanced training in quantitative methods, econometrics, and research design.
- Enhanced and optimized machine learning algorithms for the large-scale geolocation of polling stations across Brazilian territory using data from CNEFE, TSE, and the Rural Census.

FGV/EAESP/CEPESP – São Paulo/SP Since January 2024

Postdoctoral Researcher

- Conducting research on the impact of public policy and electoral behavior using causal identification methods on electoral (TSE) and sensitive government (RAIS) data, accessed in a secure data room.
- Responsible for the curation, management, and expansion of the CEPESP Data electoral database, ensuring data integrity and availability for the research community.
- Applying Deep Learning (Neural Networks) models to structure and classify unstructured data, creating new datasets for political analysis.

RELEVANT PROFESSIONAL EXPERIENCE

UNDP Consultant for Data Cleansing and AI Information Extraction October 2025 – June 2026
UNDP – CADE (Administrative Council for Economic Defense)

- Led the data cleansing and structuring project for CADE's historical administrative proceedings (pre-2012), processing a massive collection of over 760 voluminous case files (approx. 70 GB in PDFs) to support audits by oversight agencies, such as the TCU.
- Developed and implemented an advanced natural language processing pipeline, utilizing state-of-the-art OCR (PaddleOCR in a Docker environment) for digitization, integrated with Large Language Models (LLM – Gemini 2.5 Flash) for automated data extraction.
- Applied complex prompt engineering strategies and Map-Reduce (Chunking) architecture to overcome token limits in proceedings exceeding 13 million characters, ensuring the precise extraction of financial, legal, and chronological metadata.
- The project transformed decades of unstructured files into comprehensive relational databases (JSON/CSV) and textual archives (Markdown), enabling data cross-referencing with legacy systems (BI) and the precise traceability of convictions, payments, and Active Debt enrollments.

Analysis of Subnational Responses to the COVID-19 Pandemic April 2020 – April 2021
Latin American COVID-19 Observatory | Miami University

- As the lead for Brazil's data in an international research consortium, I collected and analyzed the policy responses of all 27 Brazilian states to the COVID-19 pandemic.
- Implemented an automated data collection system via web scraping in Python to monitor state government portals, creating a novel longitudinal database that also integrated population mobility data.
- Developed a composite index to assess the stringency and timeliness of state policies and conducted contingency analyses to map correlations between NPIs and dozens of health and socioeconomic variables.
- The main finding of the analysis, which resulted in international publications, was the identification of a correlation between the governors' political affiliation and the stringency of the implemented measures.
- The resulting database and analyses provided the evidence base for two academic publications in international journals and for a Latin American interactive dashboard.

Electoral Database Creation with Deep Learning December 2024 – February 2025
FGV | CEPESP | CEPESPData

- Built the first unified database on candidate assets in Brazil by structuring and categorizing over 20 million records from 16 years of elections (2006–2022), creating a resource that enables new research on the topic.
- Developed and trained a Deep Learning (BERT) model using supervised learning and active learning strategies to optimize and scale the data labeling process.
- Enhanced the automatic classification accuracy through fine-tuning techniques, hyperparameter tuning, and uncertainty measurement with the Breaking Ties method.

Data Manager for School Dropout Prevention October 2022 – August 2024
Cidade Escola Aprendiz | Fundação Vale | Territórios em Rede

- Led the project's data division, responsible for the data collection, management, and analysis strategy to combat school dropout in 16 municipalities across the country.
- Developed and automated previously manual cataloging and monitoring processes, ensuring greater agility, security, and quality in the project's data management.
- Built an evaluation and monitoring system for over 10,000 children, processing monthly data from school questionnaires to generate insights on participant progress.
- The generated analyses and sociodemographic data directly supported field teams in searching for and monitoring out-of-school children, as well as measuring the project's impact on the communities.

RELEVANT SCIENTIFIC PUBLICATIONS

- TOUCHTON, KNAUL, ARREOLA-ORNELAS, **FAGANELLO** et al. "Non-Pharmaceutical Interventions to Combat COVID-19 in the Americas Described through Daily Sub-National Data". *Scientific Data* 10, n. 1 (2023): 734. https://doi.org/10.1038/s41597-023-02638-6.
- **FAGANELLO**, SIMONI JR., CATELANO. "Revisitando 1989: uma análise da eleição de Collor com novos dados e modelos de regressão espacial". *Revista de Sociologia e Política* 30 (setembro de 2022): e013. https://doi.org/10.1590/1678-98732230e013.
- CEPALUNI, CHEWNING, DRISCOLL, **FAGANELLO**. "Conditional cash transfers and child labor". *World Development* 152 (abril de 2022): 105768. https://doi.org/10.1016/j.worlddev.2021.105768.
- TOUCHTON, KNAUL, ARREOLA-ORNELAS, **FAGANELLO** et al. "A Partisan Pandemic: State Government Public Health Policies to Combat COVID-19 in Brazil". *BMJ Global Health* 6, n. 6 (2021). https://doi.org/10.1136/bmjgh-2021-005223.

OTHER PROFESSIONAL EXPERIENCES

Data Consulting for a Dissertation on Distance and Electoral Abstention May – June 2026
FGV | Professional Master's in Public Management and Policy

- Led the design and construction of the dataset for a professional master's dissertation, measuring the distance between voters and polling stations across the state of São Paulo (a panel of roughly 34,000 polling stations, 10 elections, 2006–2024) via geocoding and road-network routing (OpenStreetMap/dodgr).
- Applied fixed-effects regression models (pooled OLS) controlling for neighborhood-level socioeconomic variables – income, education, race, traditional communities, informal settlements – extracted through geoprocessing of Census/IBGE data, identifying a robust association between distance to the polling station and abstention.
- Conducted complementary analyses (income and abstention via ESEB microdata; corroboration with a TRE-SP voter satisfaction survey) and produced complete technical reports, with robustness diagnostics, to support the dissertation writing.

Data Lead for a Large-Scale Field Survey (+3000 Participants) July – August 2025
DIEESE | F&F Consultoria

- Led the data strategy and execution for a complex field survey with over 3000 street vendors in São Paulo, ensuring the project's completion with agility, security, and on schedule.
- Developed the end-to-end monitoring system, including programming the questionnaire (xlsform) with automated selection rules and building dashboards for real-time, geolocated tracking of field interviews.
- Using R and automation tools, I managed the data processing pipeline, ensuring the delivery of a final, clean, and auditable database.

Real Estate Prospecting System with Web Scraping April 2025
Private Client | Real Estate Agency

- Built and deployed a client-facing application for data prospecting and analysis in the real estate sector, powered by a Web Scraping algorithm.
- The core of the solution is a custom Python scraper that employs browser automation and advanced evasion techniques to bypass anti-bot defenses.
- The tool automated asset prospecting, providing strategic data that optimized the real estate agency's buying and selling decisions.

Analysis of the Impact of Electoral Management Bodies on Democratic Quality April – May 2025
University of Notre Dame

- In an international project with researchers from the University of Notre Dame, I structured and developed a global and comparative database on Electoral Management Bodies (EMBs), managing the data collection, cataloging, and organization.
- Using R programming, I conducted the initial statistical analyses to model and evaluate the impact of EMBs' characteristics on the quality and integrity of elections on a global scale.

Geolocation of Catholic Churches in Brazil with Machine Learning August – October 2024
University of Texas at Austin

- In collaboration with the University of Texas at Austin, I built a novel georeferenced database to map the location of Catholic churches throughout Brazilian territory. This database provided the foundation for a new line of research on Brazilian politics within the university.
- The project involved processing raw data from the National Registry of Addresses for Statistical Purposes (CNEFE) and applying a Machine Learning algorithm for the precise identification of the institutions, with the final database delivered in RData and CSV formats.

Research Data Audit and Validation Consultancy July – October 2023
IPSOS – Mozambique

- Acting as a consultant, I conducted a complete audit of the data pipeline for a survey with small merchants in Mozambique, reviewing all data cleaning and analysis codes originally developed in SPSS.
- I migrated the entire analysis routine from SPSS to R, a process that identified and corrected inconsistencies in the previous results, ensuring the validity and reliability of the data for the client.

Consultancy on Democratic Adherence and Partisan Identity June – July 2023
CESOP – Center for Public Opinion Studies | UNICAMP

- Acting as a consultant for CESOP, I conducted an in-depth analysis of the ESEB 2022 (Brazilian Electoral Study) to identify the profiles of democratic adherence and partisan identity in Brazil.
- I applied inferential statistical tests with p-value simulation (Monte Carlo) and adjusted residuals analysis to determine the statistical significance of associations between variables of democratic adherence, partisan identity, and social profiles.
- I developed and delivered two comprehensive reports on electorate segmentation, providing CESOP with a validated analytical basis to support research on political behavior.

Social Media Image and Crisis Management Analysis April 2023 – November 2024
Brazilian Federal Government | Nova SB

- Developed and delivered over 40 analysis reports on the federal government's image on social media, monitoring public perception on critical issues such as image crises and high-impact events (e.g., floods in RS state, PCC activities).
- Implemented a network analysis methodology using Gephi to build graphs, mapping the connections and interaction dynamics between key online actors and influencers.
- Structured a data collection and analysis pipeline that combined automated extraction for rapid delivery with qualitative adjustments to ensure the high quality and accuracy of the insights.
- The generated analyses were used directly by the government for crisis management and for the strategic monitoring of the electorate's opinion.

Data Engineering for Research on Violence in Rio de Janeiro December 2023
FGV

- Contracted by FGV researchers to develop a solution to overcome the difficulty of extracting homicide data (2014–2022) from a restrictive Tableau dashboard.
- Designed and implemented a hybrid web scraping methodology, combining browser automation with Selenium and the specialized tableauscraper package to access and extract individual data points.
- The work resulted in the creation of a novel geolocated database, enabling the analysis of large-scale homicide patterns in Rio de Janeiro for the first time.
- The data extraction solution enabled the continuation of a research project on violence at FGV that had been stalled by technical barriers to data access.

Workshop: Electoral Maps with R November 2023
FGV | CEPESP – São Paulo/SP

- Co-taught a technical workshop on the analysis and visualization of electoral geospatial data with R, in partnership with Lucas Gelape and João V. Guedes-Neto (FGV CEPESP), aimed at researchers and data science professionals.
- The content covered the characteristics of spatial data, linking electoral results to space (including intramunicipal analyses), and producing maps focused on electoral geography.
- Course materials are available at: gv-cepesp.github.io/mapaseleitorais2023.

Geospatial Mapping Platform for Electoral Campaigns 2022 – 2024
VotoMaps

- Conceived, developed, and delivered an online platform of electoral reports and maps for over 40 candidates for state and city representative in the 2022 and 2024 elections.
- Programmed the interactive maps using JavaScript, HTML, and the Mapbox library, with features for geolocalized visualization of votes by polling station and identification of electoral strongholds.
- Implemented an electorate profiling feature by cross-referencing voting data with information from IBGE's

census tracts to generate demographic insights.
- The platform was used directly by campaign teams to inform and guide strategies for resource allocation and electoral mobilization.

Quantitative Consultancy on School Dropout and Child Labor March – September 2022
Cidade Escola Aprendiz | Fundação Vale

- Acting as a statistician/quantitative specialist, I conducted an analysis of the pandemic's impact on school dropout in the city of São Paulo using secondary data from the School Census.
- I performed data prospecting and descriptive and spatial statistical analysis, identifying a sharp drop in enrollment numbers, especially in public schools in high-vulnerability areas.
- I developed the sampling plan for primary data collection and modeled the obtained quantitative data, generating reports and cartographic representations (maps) for the project.
- The analyses and reports produced were used by the city's government to inform the reformulation of educational policies.

Data Intelligence Consultancy for Federal Deputy Campaign April – September 2022
Electoral Campaign of Federal Deputy Candidate Gabriela Manssur

- Conducted quantitative and geocoded data analysis to inform the strategy of a Federal Deputy campaign.
- Developed a geographic area prioritization model, identifying electoral strongholds with the highest engagement potential for the candidate, whose main platform was the fight against violence against women.
- The data analyses and geospatial assessments directly guided advertising investment decisions and the allocation of field campaign resources.

Geographic and Strategic Analysis for Electoral Campaign February 2022
Electoral Campaign | Wagner Romão – PT – Campinas 2022

- Developed a geographic analysis of a candidate's voting patterns in the 2018 and 2020 elections to identify opportunities and map electoral strongholds for the 2022 campaign.
- Created interactive maps and online reports using JavaScript and Mapbox, cross-referencing voting data with information from census tracts (IBGE) to create a demographic profile of the electorate in key territories.
- The mappings and profile analyses were used directly by the campaign team to guide campaign strategy, optimizing resource allocation and field mobilization.

Pioneering Study of Law Enforcement Candidacies in São Paulo October – December 2021
University of Bradford

- In a research project with the University of Bradford (England), I conducted the first in-depth study on the candidacy patterns of security agents (civil police, military police, and private security guards) in São Paulo.
- I conducted an electoral and territorial analysis spanning 16 years of municipal elections (2004–2020), identifying and comparing the sociodemographic and voting profiles of the different careers.
- The generated database and analyses served as the foundation for an international research project on the impact of the politicization of police forces in Brazilian electoral campaigns.

Campaign Diagnostics and Strategy (PREVI/BB Council) November – December 2021
Grupo Mais | PREVI | Banco do Brasil

- Conducted a mixed-methodology research study (qualitative and quantitative) with Banco do Brasil employees to develop the strategy for an electoral campaign for the PREVI council, one of the largest pension funds in Latin America.
- Developed the script and moderated online focus group sessions, investigating employee perceptions and demands regarding the pension fund and the candidates.
- Performed the analysis of quantitative (online survey) and qualitative data, consolidating the findings into a comprehensive diagnostic report for the campaign team.
- The insights generated by the research directly informed the definition of the messages, platform, and communication strategies for the Grupo Mais campaign.

Android App Development for Telephone Survey Automation September – October 2021
APEOESP | DIEESE | F&F Consultoria

- Participated in a telephone opinion survey with 120 unionized teachers and simultaneously identified an opportunity to optimize the dialing process.
- Developed an Android application using Kotlin that automated telephone calls from a list of interviewees.
- The solution eliminated the need for manual dialing, reducing idle time between interviews and optimizing the interviewers' workflow.
- The application reduced the client's operational costs by replacing the need for a professional CATI (Computer-Assisted Telephone Interviewing) infrastructure.

Digital Communication Campaign Automation (Python & Selenium) August 2021
Political Party

- Developed a Python algorithm to automate communication with party members, aiming to increase engagement and registrations for the "Meeting of the Human Rights Sector".
- The solution utilized the Selenium library for browser automation, sending personalized invitations via WhatsApp and email to an authorized contact list.
- The automation optimized the event's outreach strategy, ensuring direct and efficient communication with members and boosting the number of registrations by the deadline.

Statistical Analysis for Research on Economic Behavior August 2021
Private Client

- Provided statistical consulting for a master's dissertation on the topic "The Practices of Self-Nudging Behavior (Self-Nudges) from a sociological perspective".
- Analyzed data from the online survey, developing cross-tabulations and applying significance tests to identify the most relevant response patterns.
- Developed and delivered a complete analytical report with the quantitative results, which served as the foundational basis for the dissertation's argumentation and conclusion.

Data Analysis Consultancy for Electoral Scenario March 2021
Perspectiva Comunicação | Belém-PA 2020 Elections

- Conducted a diagnostic on an unexpected electoral outcome in Belém, investigating the factors that led a candidate with no political experience to the second round.
- Performed a quantitative and geospatial data analysis (2016–2020) to map the candidate's electoral strongholds and identify the socioeconomic profile of their voters.
- Conducted interviews with specialists from the research firm IBOPE to deepen the analysis and understand why voter intention polls failed to capture the last-minute movement.
- The final diagnosis concluded that a shift in voting patterns hours before the election, subtly indicated in the data, was not fully captured by the polls, providing the client with a clear explanation for the result.

Electoral Data Intelligence Consultancy November 2020
Concordia Public Affairs

- Provided data consulting for Concordia Public Affairs, delivering a comprehensive analysis of the 2020 municipal election results.
- Developed a nationwide database consolidating the voting results for mayors and city councilors in every municipality in the country.
- Conducted an in-depth statistical analysis, generating a report with key metrics such as re-election rates, the gender profile of elected officials, and the level of fragmentation in municipal legislatures.
- The database and analytical report were used by the consultancy to provide their clients with insights into Brazil's new political-electoral landscape.

Analysis of Political Party Mentions on Twitter September 2019
Brazilian Politics Study Group (POLBRAS) | UNICAMP

- Developed a web scraping algorithm in Python to build a longitudinal database, capturing mentions of Brazilian political parties on Twitter over a two-year period.
- Implemented and managed the data collection and structuring process, creating a robust dataset for analyzing the performance and perception of parties on social media.
- The generated database served as a primary data source to support the research of the Brazilian politics study group at Unicamp.

Telephone Survey Automation for a Labor Union (Kotlin/Android) July – August 2019
SINDMETALSJC | DIEESE | F&F Consultoria

- While working on the field team for an opinion survey with members of the Metalworkers' Union, I proposed and developed a solution to automate the telephone call process.
- I programmed a native Android application in Kotlin that automatically dialed interviewees from a predefined list, managing the call flow.
- The tool increased the productivity of the interviewer team by eliminating manual dialing and reducing idle time between calls.

Database Construction on Slave Labor in Brazil June – August 2019
FGV

- Developed and structured a comprehensive database on slave labor in Brazil to support a research study by the Fundação Getúlio Vargas.
- The project involved the collection and consolidation of multiple sources, including public data from the Ministry of Labor and novel datasets on enforcement actions.
- Analyzed sensitive government data in a secure data room to enrich the database, ensuring the confidentiality and security of the information.
- The consolidated database served as the primary source of evidence for the FGV's research, enabling in-depth analyses on the topic in the country.

Geographic Analysis for a City Councilor Campaign March 2019
Electoral Campaign | Wagner Romão – PT – Campinas 2016

- Conducted a geographic analysis of voting patterns from previous municipal elections to map strongholds and identify territories with growth potential for the 2016 campaign.
- Created interactive maps using JavaScript and Mapbox, cross-referencing voting data by polling precinct with information from census tracts (IBGE) to analyze the electorate's socioeconomic profile.
- The mappings and profile analysis guided the field mobilization strategy, allowing the campaign to focus its resources and efforts on priority areas of the city of Campinas.

Large-Scale Web Scraping for a Research Funding Audit August – September 2018
ADUNICAMP | UNICAMP

- Contracted by ADUNICAMP to investigate and quantify the perceived decline in FAPESP research funding for UNICAMP over the years.
- Developed a web scraping algorithm in Python using Scrapy that collected and structured over 236,000 research grants from the FAPESP website (1992–2017), storing the data on a MongoDB server.

- Conducted a statistical analysis of the data in R, identifying a 20.5% drop in total grants for UNICAMP and a 47.3% reduction (an estimated R$18 million) in scholarship payments between 2013 and 2017.
- Prepared the final research report, which provided ADUNICAMP with concrete evidence to inform policy formulation and the defense of research at the university.